

PROPOSAL

"SMART QUIZ MANAGEMENT AND PERFORMANCE ANALYSIS SYSTEM"

NAME: MALIK HUSSAIN ALI HAIDER

REG NO: F-24-BSSE-5A-020

INSTRUCTOR: MR.WAQAS AHMED

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1. Introduction:

The **Smart Quiz Management and Performance Analysis System** is a web-based platform for students to attempt quizzes and for instructors to manage questions and analyze student performance. Unlike a basic quiz system, it will identify strong and weak topics and provide recommendations for further practice.

The system will include subject/topic management, timed and randomized quizzes, automatic evaluation, quiz history, performance analysis, recommendations, and role-based access for students, instructors, and administrators.

2. Application of CCP Principles

2.1 Depth of Knowledge Required:

The project requires knowledge of web development, databases, algorithms, performance analysis, question selection, and decision-making logic. The system will analyze quiz results to identify weak areas and recommend suitable practice.

2.2 Conflicting Requirements:

The system must balance:

- Personalization with fair assessment
- Security with ease of use
- Quiz difficulty with student engagement

2.3 Depth of Analysis Required:

Different methods can be used to calculate performance, identify weak topics, and recommend questions. The system will consider factors such as accuracy, previous results, topic performance, and question difficulty to select a suitable approach.

2.4 Familiarity of Issues:

Although online quizzes are common, this project extends the basic concept by connecting assessment results with topic-wise analysis and personalized practice recommendations.

2.5 Extent of Stakeholder Involvement:

The main stakeholders are:

- Students: attempt quizzes, view results, track progress, and receive recommendations.
- Instructors: create quizzes, manage questions, and analyze performance.
- Administrators: manage users, data, and access permissions.

2.6 Interdependence:

The system's components are interconnected. Quiz attempts produce results, results are stored in the database, performance data is analyzed, and the analysis is used to generate recommendations.

2.7 Consequences:

Incorrect scoring or analysis may provide students with misleading feedback. Unauthorized access may also create privacy issues. Therefore, the system requires accurate scoring, reliable data processing, and proper access control.

3. Proposed System and Objectives:

The main objective is to develop a quiz platform that conducts assessments and provides meaningful performance feedback.

Objectives:

- Provide quiz creation and attempting functionality.
- Automatically evaluate objective questions.
- Analyze overall and topic-wise performance.
- Identify weak areas.
- Recommend suitable practice quizzes/questions.
- Maintain quiz history and performance records.
- Provide role-based access for different users.

4. Planned Features:

- Student registration and login
- Role-based access
- Subject and topic categorization
- Quiz and question management
- Multiple-choice questions
- Timed and randomized quizzes
- Automatic scoring
- Quiz history
- Topic-wise performance analysis
- Weak-topic identification
- Recommended practice
- Student and instructor dashboards
- Basic leaderboard
- Secure data storage

5. Technical Approach:

The project will use a web-based client-server architecture. The frontend will provide user interfaces, while the backend will handle authentication, quiz processing, scoring, analysis, and database communication.

Technologies:

- HTML, CSS, JavaScript
- React
- Node.js
- MongoDB
- Git/GitHub
- In future deploying the website

The implementation may be refined according to project requirements and instructor feedback.

6. Expected Outcome:

The expected outcome is a functional quiz system that allows students to attempt quizzes, receive results, monitor performance, identify weak topics, and receive practice recommendations. Instructors will be able to manage quizzes and questions and view student performance.

7. Conclusion:

The Smart Quiz Management and Performance Analysis System will provide more than basic quiz functionality by connecting assessment results with performance analysis and targeted practice. The project demonstrates CCP principles through its required computing knowledge, conflicting requirements, analysis, stakeholder involvement, interconnected components, and consequences of incorrect results or unauthorized access while maintaining a practical medium-level scope.